

HR Autonomous Agent Architecture

HR-operations-first reasoning, execution, recovery, and reflection

1. Overview

The agent is designed as a controlled autonomous workflow for HR operations. It interprets goals in business terms first, prepares the target environment, inspects relevant application logic, executes the smallest safe action, verifies the outcome, and continuously improves through recorded memory and reflection.

2. Key Features

- Business-operation-first interpretation of user goals.
- Warm-start behavior that reuses known context before broad discovery.
- Mandatory reasoning and action justification before mutation.
- Targeted system inspection to support safe execution.
- Bounded retry behavior with explicit recovery logic.
- Persistent memory, logs, and post-run reflection.

3. Design Goals

- Make HR outcomes the primary decision lens before treating a request as an engineering task.
- Start quickly by using configured startup data and warm context by default.
- Prefer narrow, reversible operations with verification after every meaningful step.
- Escalate rather than guess when confidence or evidence is insufficient.
- Retain lessons across runs so later execution becomes safer and faster.

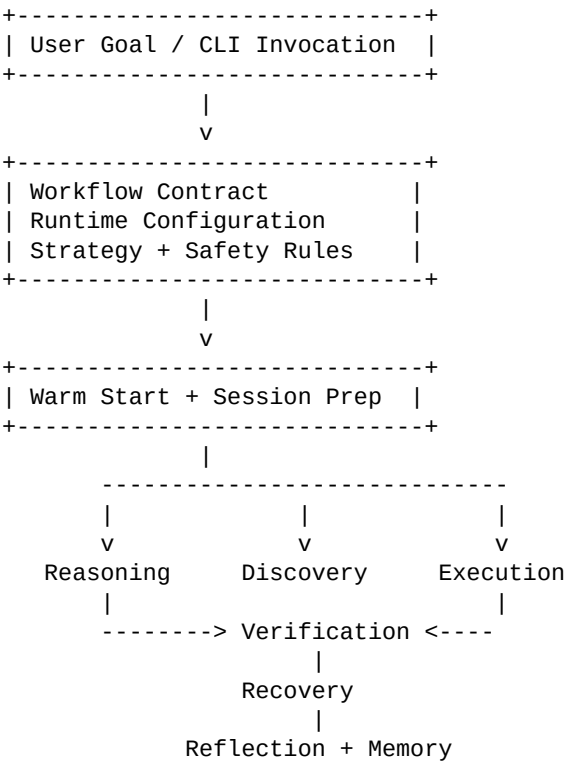
4. Core Operating Principles

- Summary-only inspection output during normal operation.
- Targeted discovery before broad rediscovery.
- Read-first validation before write actions when possible.
- Reflection after every completed or terminally failed run.

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System structure and core components

5. System Architecture



6. Component Responsibilities

Component	Responsibility
Workflow Layer	Defines the governing contract for runtime behavior.
Config Layer	Supplies startup behavior, environment assumptions, defaults, and safety settings.
Reasoning Agent	Interprets intent and produces the safest execution path.
Discovery Agent	Refreshes only the technical facts needed for the active goal.
Execution Agent	Performs one justified action at a time and verifies the outcome.
Recovery Agent	Diagnoses failures, applies narrow corrections, and retries only when justified.
Reflection Agent	Evaluates results and improves future heuristics.
State & Memory	Stores workflow progress, history, and operational evidence.

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Runtime flow and decision behavior

7. Runtime Workflow

```
Receive Goal
|
Load Workflow Contract and Runtime Settings
|
Load Warm State and Known Context
|
Validate Service Availability and Session State
|
Reasoning Agent: classify goal and choose safest path
|
Discovery Agent: inspect only the modules needed now
|
Execution Agent: perform smallest justified action
|
Verification: confirm response and state change
|
Recovery Agent: retry only if failure is recoverable
|
Reflection Agent: store lessons for future runs
```

8. Confidence Model

- High confidence: proceed with normal safeguards because the business meaning, module, and action path are all supported.
- Medium confidence: proceed only with narrow and reversible actions plus explicit verification.
- Low confidence: inspect further or escalate instead of guessing.

9. Example Reasoning Snapshot

```
Goal: approve pending attendance requests for last week
Task Type: attendance operations
Confidence: Medium
Relevant Areas: attendance, approval flow, authentication
Inferred Action Path: authenticate -> confirm scope ->
                        approve eligible requests -> verify

Strategy:
1. Validate configured backend and active session
2. Confirm attendance approval workflow
3. Execute the smallest safe approval action
4. Verify the post-action state
5. Record the result, memory update, and reflection
```

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Configured profile, guardrails, and completion guarantees

10. Configured Operating Profile

Setting	Value
Entry Command	agent run "<goal>"
Default Persona	hr operations specialist
Goal Interpretation	business-operation-first
Startup Mode	configured-path warm-start
Runtime Shell	powershell
Warm Cache Reuse	enabled when the active target still matches
Discovery Preference	targeted refresh before any broad rediscovery
Authentication	required before task execution
Retry Limit	3
Safe Mode	enabled
Bulk Operations	disabled unless reasoning marks them explicitly safe
CLI Visibility	summary-only during inspection
Action Reasoning	required before every execution step
Memory Update	after each task and major state transition
Reflection	required after completion or terminal failure

11. Risk and Recovery Guardrails

- Prefer read-only verification before any write action.
- Avoid broad or bulk changes unless the reasoning step explicitly justifies them.
- Retry only when the failure is plausibly recoverable and something meaningful changes.
- Escalate when evidence is conflicting, destructive side effects appear, or confidence remains too low.

12. Completion Guarantees

- A run ends only after it reaches a terminal outcome such as completed, failed, or escalated.
- Execution history, failure records, state tracking, task memory, and reflection must all be updated.
- Recorded lessons are expected to influence later runs so the system becomes safer and more consistent over time.